Building a Reliable and Repeatable Health Data Pipeline for Hospital Decision-Making

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1. Executive Summary

In hospital settings, leaders depend on timely, consistent data to make operational decisions. However, manual and ad-hoc data preparation can introduce hidden issues—such as inconsistent logic, undocumented transformations, and limited traceability—that reduce trust in results.

This project demonstrates a small, end-to-end health data pipeline that ingests CSV, JSON, and HTML sources, applies standardized cleaning and validation, and generates reproducible visuals. The goal is to show a transparent, repeatable workflow that supports reliable reporting and stakeholder confidence.

2. Pipeline Goals and Operational Value

This pipeline standardizes data processing across reporting workflows to improve reliability and repeatability. It reduces inconsistency risk by using validated, rerunnable code rather than manual steps. It also surfaces data quality issues early and documents key processing decisions so stakeholders can understand how results were produced.

3. Data pipeline Design and Implementation

3.1 Data Sources and Ingestion (Step 1)

The pipeline begins by ingesting three public health data sources that represent common formats used in hospital analytics. A CSV dataset containing age-adjusted death rates is loaded using pandas.read_csv(). A JSON dataset is retrieved from a CDC API endpoint using pandas.read_json(). An HTML table is extracted from a CDC data brief webpage using pandas.read_html(). All sources are read without modification to preserve traceability.

3.2 Initial Exploration and Understanding (Step 2)

After ingestion, the pipeline performs initial exploration using functions such as head(), info(), and describe(). This step confirms schema, data types, and value ranges, and helps identify structural differences across sources early in the process.

3.3 Cleaning, Standardization, and Reshaping (Step 3)

Once the structure of each dataset is understood, targeted cleaning steps are applied. Relevant columns are selected, numeric values are coerced to appropriate types, and duplicate records are removed where necessary. A minimal reshaping step using melt() is demonstrated on the CSV dataset to prepare the data for reporting-style analysis and visualization.

3.4 Data Validation and Quality Checks (Step 4)

Explicit data validation is a core component of this project. The most complete validation checks are applied to the CSV dataset, which represents the primary reporting table.

These checks include missing value detection, duplicate identification, distribution analysis using histograms, outlier detection using an IQR-based approach, logical checks for impossible values, and exception handling to prevent pipeline failure.

For the JSON and HTML datasets, the pipeline focuses on structural validation, type conversion, and stable visualization. In a production environment, the same validation logic could be extended to these sources if they were used for primary reporting.

3.5 Visualization and Review (Step 5)

After validation, the pipeline produces stakeholder-friendly visualizations for each data source. These charts summarize key patterns and serve as a final visual quality check.

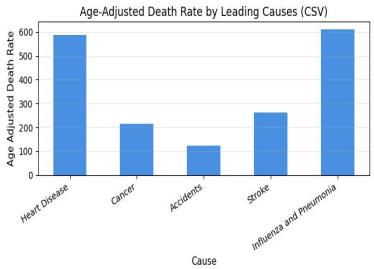

Figure 1

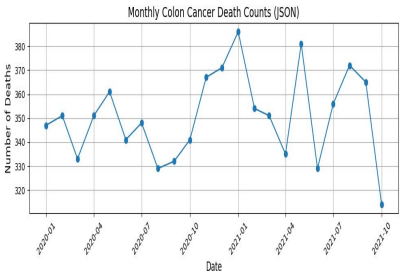

Figure 2

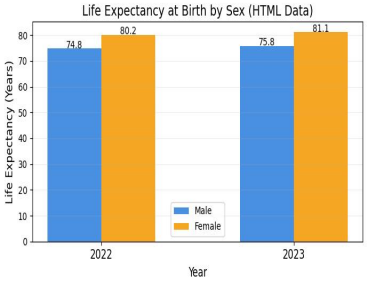

Figure 3

4. Data Quality Status

Based on validation checks and visual review, the overall data quality is suitable for analysis and reporting. The primary CSV dataset shows minimal missing values, no unresolved duplicates, and no logically impossible values. Outliers identified through statistical checks are surfaced for review rather than automatically removed.

5. Risks and Limitations

Potential risks include changes to upstream data structure, evolving metric definitions, and the need for clinical or operational context when interpreting extreme values. These risks are mitigated through versioned inputs, explicit validation, and clear documentation.

6. Data Governance and Accountability

Governance practices are embedded throughout the pipeline. Raw inputs are preserved to maintain traceability, cleaned outputs are versioned, and validation results are documented with each run. Clear accountability for data ownership and maintenance supports audit readiness and evolving compliance expectations in healthcare analytics environments.

7. Conclusion and Next Steps

This project demonstrates how a well-documented data pipeline can improve reliability, transparency, and efficiency in hospital analytics workflows. Future enhancements may include automated executions, validation alerts, and integration with dashboards as organizational reporting needs grow.